

Gavin Kerrigan

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Research

Postdoctoral Research Assistant in Machine Learning

Grade 8 Jan. 2026 - Present
Grade 7 Feb. 2025 - Jan. 2026
University of Oxford, Department of Statistics
RainML Lab (PI: Tom Rainforth)

Graduate Student Researcher

Sep. 2019 - Dec. 2024

University of California, Irvine, Department of Computer Science
DataLab (PI: Padhraic Smyth)

Undergraduate Research Assistant

Sep. 2018 - June 2019

The Pennsylvania State University, Department of Computer Science

Education

Ph.D., Computer Science

Sep. 2019 - Dec. 2024

University of California, Irvine GPA: 4.0/4.0
Thesis: Infinite-Dimensional Generative Models through the Transport of Measure
Advisor: Padhraic Smyth

B.Sc., Mathematics

2015 - 2019

The Pennsylvania State University, Schreyer Honors College GPA: 3.98/4.0
Minor: Physics

Teaching

Tutor

Jan. 2026 - Mar. 2026

SB2.2: Statistical Machine Learning, University of Oxford

Instructor

March 2023 - June 2023

CS 178: Machine Learning and Data Mining, UC Irvine
■ Enrollment: 162

Head Teaching Assistant

Sep. 2022 - Dec. 2022

CS 178: Machine Learning and Data Mining, UC Irvine
■ Managed a team of 5 graduate TAs in a course with an enrollment of 272
■ Developed homework assignments, weekly discussions, exams, and rubrics

Guest Lecturer

Fundamentals of AI II: Modern Statistical Concepts, University of Oxford Nov. 2025
CS 274E: Deep Generative Models, UC Irvine Nov. 2022, Nov. 2023, Nov. 2024

Private Tutor

Jul. 2019 - Dec. 2024

Machine learning, mathematics, and statistics

- Perfect (5.0/5.0) rating across more than 300 hours of tutoring and 200 reviews

Mathematics Tutor and Peer Coordinator

Jan. 2017 - June 2019

Penn State Learning

- Tutored undergraduate mathematics courses in one-on-one and small group settings
- Trained, supervised, and interviewed a team of more than 60 undergraduate tutors

Publications

Google Scholar: <https://scholar.google.com/citations?user=2F2XCy8AAAAJ>

* denotes joint authorship

Conference

1. **G. Kerrigan**, K. Nelson, P. Smyth, 'EventFlow: Forecasting Continuous-Time Event Data with Flow Matching,' *Proceedings of the 29th International Conference on AI and Statistics (AISTATS 2026)*, to appear, May 2026 (**Oral, top 0.5%**).
2. **G. Kerrigan**, C. Naesseth, T. Rainforth, 'A Geometric Approach to Optimal Experimental Design,' *Proceedings of the 29th International Conference on AI and Statistics (AISTATS 2026)*, to appear, May 2026.
3. A. Saravanan, R. Knowles, **G. Kerrigan**, T. Rainforth, 'Scaling Bayesian Experimental Design to High-Dimensions with Information-Guided Diffusion,' *International Conference on Learning Representations (ICLR 2026)*, to appear, April 2026.
4. A. Mammadov, J. Yao, J. Berner, **G. Kerrigan**, J. C. Ye, K. Azizzadenesheli, A. Anandkumar, 'Guided Diffusion Sampling on Function Spaces with Applications to PDEs,' *Advances in Neural Information Processing Systems 39 (NeurIPS 2025)*, December 2025.
5. **G. Kerrigan**, G. Migliorini, P. Smyth, 'Dynamic Conditional Optimal Transport through Simulation-Free Flows,' *Advances in Neural Information Processing Systems 38 (NeurIPS 2024)*, pp.93602–93642, December 2024.
6. P. Srivastava, R. Yang, **G. Kerrigan**, G. Dresdner, J. McGibbon, C. Bretherton, S. Mandt, 'Precipitation Downscaling with Spatiotemporal Video Diffusion,' *Advances in Neural Information Processing Systems 38 (NeurIPS 2024)*, pp. 56374-56400, December 2024.
7. **G. Kerrigan**, G. Migliorini, P. Smyth, 'Functional Flow Matching,' *Proceedings of the 27th International Conference on AI and Statistics (AISTATS)*, Proceedings of Machine Learning Research, PMLR 238:3934-3942, April 2024 (**Oral, Outstanding Student Paper Highlight award (7/2181)**).
8. **G. Kerrigan**, J. Ley, P. Smyth, 'Diffusion Generative Models in Infinite Dimensions,' *Proceedings of the 26th International Conference on AI and Statistics (AISTATS)*, Proceedings of Machine Learning Research, PMLR 206:9538–9563, April 2023.
9. **G. Kerrigan**, P. Smyth, and M. Steyvers, 'Combining Human Predictions with Model Probabilities via Confusion Matrices and Calibration,' *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, pp.4421–4434, December 2021.

Journal

10. C. Guilloteau, **G. Kerrigan**, K. Nelson, G. Migliorini, P. Smyth, R. Li, E. Foufoula-Georgiou, 'A Generative Diffusion Model for Probabilistic Ensembles of Precipitation Maps Conditioned on Multisensor Satellite Observations,' *IEEE Transactions on Geoscience and Remote Sensing*, doi:10.1109/TRGS.2025.3548518, March 2025.
11. M. Steyvers, H. Tejada, **G. Kerrigan**, P. Smyth, 'Bayesian Modeling of Human-AI Complementarity,' *Proceedings of the National Academy of Sciences*, 19(11):1-7, March 2022.

Workshop

12. **G. Kerrigan**, G. Migliorini, P. Smyth, 'Functional Flow Matching,' *NeurIPS Workshop on Diffusion Models*, 2023.
13. **G. Kerrigan**^{*}, D. Slack^{*}, J. Tuyls^{*}, 'Differentially Private Language Models Benefit from Public Pre-Training,' *PrivateNLP: EMNLP Workshop on Privacy and Natural Language Processing*, pp.39-45, 2020.

Under Review

14. A. Forster, **G. Kerrigan**, D. R. Ivanova, T. Rainforth, 'Improving Robustness to Model Misspecification in Bayesian Experimental Design,' *under review*, January 2026.

Honors and Awards

Research

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| Oral Presentation (0.5% of submissions), AISTATS | 2026 |
| Outstanding Student Paper Highlight award, AISTATS | 2024 |

University of California, Irvine

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| Hasso Plattner Institute Fellowship | 2021 - 2024 |
| Graduate Dean's Recruitment Fellowship | 2019 |

The Pennsylvania State University

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| Steven and Sherry McCrystal Mathematics Award | 2019 |
| Mary Lister McCammon Scholarship | 2017, 2018 |
| Phi Beta Kappa Honors Society | 2017 |
| Hay Memorial Scholarship | 2016 |
| Mahle Trustee Scholarship in Science | 2015 |

Invited Talks

Optimal Transport and Optimal Experimental Design

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| Computational Methods in Applied Mathematics (<i>Upcoming</i>) | July 2026 |
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Generative Models and the Transport of Measure

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| Anandkumar Group, Caltech | August 2024 |
| University of Southern California (USC) | October 2024 |
| Yu Group, University of California San Diego (UCSD) | October 2024 |

Diffusion Generative Models: Methods and Applications

NASA Jet Propulsion Laboratory (JPL)

June 2024

Functional Flow Matching

The 27th International Conference on AI and Statistics (AISTATS)

May 2024

Deep Generative Models in Infinite-Dimensional Spaces

Center for Machine Learning Seminar, UC Irvine

Jan. 2024

Probability and Combinatorics Seminar, UC Irvine

Nov. 2023

AMLab Seminar, University of Amsterdam

Nov. 2023

A Tutorial on Diffusion Generative Models

Hasso Plattner Institute

Sep. 2023

Service

Departmental Committee Postdoctoral Representative

University of Oxford, Department of Statistics

Nov. 2025 - Present

Student Success Mentor

The Mentor Collective, UC Irvine Next Gen Pathways

2024

Workflow Chair

AISTATS

2024

Session Chair

AISTATS

2024

Area Chair

ICLR, ICML

2025

AI Faculty Search Committee, Student Member

UC Irvine

2021 - 2023

Machine Learning Hackathon Mentor

UC Irvine

2021

Academic Reviewing

JMLR, ICML, NeurIPS, AAAI, AABI, AISTATS, TAG-DS, The American Statistician

2025

JMLR, ICLR, CVPR, ICML, UAI, AAAI, AISTATS, NeurReps Workshop

2024

NeurIPS, AISTATS, Workshop on Deep Generative Models for Health,

2023

Workshop on Topology, Algebra, and Geometry in Data Science,

Workshop on Diffusion Models

ICML, Workshop on Geometrical and Topological Representation Learning,

2022

Workshop on Human-Machine Collaboration and Teaming

Workshop on Uncertainty in Deep Learning

2020